INCH - POUND
MIL-DTL-43124D
1 June 2018
SUPERSEDING
MIL-T-43124C
28 September 1984

DETAIL SPECIFICATION

TRUCK, HAND, PLATFORM, LAUNDRY AND DRYCLEANING, SPECIAL DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This document covers hand platform trucks of magnesium or aluminum construction and rated at 2000-pounds capacity. Running gear consists of two rigid casters and two swivel casters

1.2 Classification. Hand platform trucks should be of the following types as specified (see 6.2).

Type I - Shipping
Type II - Receiving

2. APPLICABLE DOCUMENTS

2.1 Government Documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

STANDARDS

FEDERAL

FED–STD-601 - Rubber : Sampling and Testing

MILITARY

MIL-STD-130 - Identification Marking of US Military Property

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support - Industrial Hardware Division (ATTN: Code FHTE), 700 Robbins Avenue, Philadelphia, PA 19111-5096 or email trpsptspecspa@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

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2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

ASTM International

ASTM D2287 - Nonrigid Vinyl Chloride Polymer and Copolymer Molding and Extrusion Compounds.

(Copies of these documents are available from http://www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

ASME (AMERICAN SOCIETY OF MECHANICAL ENGINEERS)

ASME-B56.11.1 - Double Race or Bi-Level Swivel and Rigid Industrial Casters

(Copies of these documents are available from https://www.asme.org or ASME, Two Park Avenue, New York, NY 10016-5990)

American Society for Quality (ASQ):

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes..

(Copies of this document are available online from http://asq.org or from the ASQ Distribution Center, 5131 S. Third Street, Milwaukee, Wisconsin 53207-6028.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2, 6.2, 6.3).

3.2 Material. The materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document.

3.2.1 Bumpers

3.2.1.1 Bumper, rubber. The rubber bumper shall be molded from commercial type nonmarking gray rubber compound having a Shore A Durometer hardness reading of 70 plus or minus 5. Rubber bumpers shall be considered nonmarking if only a slight mark is made on white bond paper which can be easily rubbed off with dry fingers. The rubber bumper shall be reinforced by a steel insert. Testing shall be as specified in 4.3.2.

3.2.1.2 Bumper, vinyl. The vinyl bumper shall be in accordance with type I, grade 15 of ASTM D2287.

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3.2.2 Casters. Casters shall conform to Group A, type I and II, class 4, style 2 of ASME-B56.11.1 The wheel diameter shall be not less than 8 inches.

3.2.3 Fastening devices. Fastening devices such as bolts, nuts, and lock washers shall be fabricated from steel and shall be plated in accordance with commercial practice.

3.2.4 Rivets. Rivets shall be fabricated from aluminum alloy.

3.3 Design and construction.

3.3.1 Type I truck. The type I truck shall consist of a deck assembly with push bar sockets, caster mounting plates, two rigid casters, two swivel casters, bumper assembly, and a push handle. The type I truck capacity and physical dimensions shall be as specified in table I.

TABLE I. Truck capacity and physical dimensions.

		Dimensions (inches)'			
	Rated capacity (pounds, minimum)	Length (overall)	Width (overall)	Height (overall)	Platform (height)
Type I	2000	60-61	42-43	39-39	12-14
Type II	2000	72-73	48-49	63-64 1/	12-14

1/ A maximum height of 71 inches is permissible on one end when used as a push bar.

3.3.1.1 Deck assembly. The frame and deck may be fabricated from one sheet with edges turned down and inward to form a channel frame and deck; or may be of one sheet (deck) assembled to a frame of extruded or formed channels (frame), or the deck may be of sectionalized construction consisting of sections in the form of extruded planks and shall interlock together to form a smooth deck which in turn shall be assembled to extruded shapes forming the frame. The corners of the deck and frame shall be formed to accept the push bar sockets specified in 3.3.1.2. The frame and deck shall be properly reinforced and shall be of all magnesium alloy or aluminum alloy fabrication. The complete assembly, including sockets, shall form a rigid structure. The deck surface, using planks, shall be smooth with joint spacings no greater than .031 (1/32) inch.

3.3.1.2 Sockets. Sockets for push bar legs shall be fabricated from magnesium alloy or aluminum alloy of cast or formed construction. Sockets shall be permanently affixed, one each, at each corner of the deck to form round corners. The top surface of the socket and deck shall be flush. The socket shall act as a stop and as a rigid support for the push bar leg. The socket hole shall be between .020 to .070 inches larger than the outside diameter of the push bar leg. Center-line span of each pair of socket holes shall not vary by more than .250 (1/4) inch.

3.3.1.3 Caster mounting plates. Caster mounting plates shall be fabricated from magnesium alloy or aluminum alloy. Caster supports may be of plate, channel, angle or combination of construction, or they may be an integral part of the underside of the sectionalized plank deck. The caster supports shall be welded, brazed or riveted, as required to the underside of the deck.

3.3.1.4 Bar, push. The push bar shall be fabricated of magnesium or aluminum alloy tubing between 1.375 (1-3/8) – 1.750 (1-3/4) inches outside diameter. The push bar shall be one piece and shall be U-shaped. The bottom of each leg may have a solid insert. When assembled to the sockets (widthwise), the handle end or top portion of push bar shall extend outward between 3 -4 inches toward the operator. The push bar shall be removable.

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3.3.1.5 Running gear. Running gear shall consist of two rigid casters and two swivel casters and shall be as specified in 3.2.2. The casters shall be bolted to the caster mounting plates by fasteners specified in 3.2.3. The casters shall be positioned so that with rated load equally distributed over the entire deck surface, the load division for each caster shall be equal to within plus or minus twenty-five (25) pounds.

3.3.1.6 Bumper assembly. Bumper assembly may consist of bumper frame supports, bumper frame and bumper. Bumper frame supports may be brackets or spacers (of magnesium or aluminum alloy) welded to and extending out from the truck frame or they may be extensions of the truck frame reinforcing members (see 3.3.1.1). Bumper frame may be rigidly fastened to the bumper frame supports or to the truck frame and shall encircle the truck frame. The bumper frame shall be fabricated of magnesium or aluminum alloy channel. Bumper shall be continuous and of material specified in 3.2.1. Bumper shall completely encircle the bumper frame and shall be securely affixed to the bumper frame by means of fasteners specified in 3.2.3 or 3.2.4. Bumpers shall be capable of being removed and replaced. Bumper shall extend outward beyond the truck frame no more than 4 inches.

3.3.2 Type II truck. Type II truck shall consist of a deck assembly, bumper assembly, caster mounting plates, two rigid casters and two swivel casters, superstructure and sockets for mounting the superstructure. Capacity and physical dimensions of truck shall be as specified in table I.

3.3.2.1 Type II, deck assembly. The type II, deck assembly shall be as specified in 3.3.1.1.

3.3.2.2 Type II, sockets. The type II, sockets shall be fabricated from magnesium alloy or aluminum alloy of cast, formed, tube, or extruded construction. Sockets shall be welded, to the truck deck to receive the rack. Sockets shall rigidly support the superstructure.

3.3.2.3 Type II, superstructure. The type II, superstructure shall consist of removable racks. The racks shall be rectangular in configuration with a “U” shaped outer member and vertical rungs spaced no more than 4.500 (4-1/2) inches apart between the “U” shaped member and a horizontal lower member. The horizontal lower member shall be located to provide legs on the end of the “U” shape member with sufficient clearance when the rack is mounted in the sockets. The end racks and the racks for one side when mounted shall have a height of not less than 63 and not more than 64 inches above the floor. The racks for the other side when mounted shall have a height of not less than 37 inches above the floor. The rack shall be made of round or square magnesium or aluminum alloy tubing. The legs of the racks may be filled with a solid metal filler for rigidity . When mounted in the sockets the top of the racks shall not sway more than .750 (3/4) of an inch from a plumb line when tested in accordance with 4.4.1.

3.3.2.4 Type II, bumper assembly. The type II, bumper assembly shall be as specified in 3.3.1.6.

3.3.2.5 Type II, caster mounting plates. The type II, caster mounting plates shall be fabricated from magnesium alloy of angle, channel or combination thereof. The caster mounting plates shall be permanently secured to the underside of the deck.

3.3.2.6 Type II, running gear. The type II, running gear shall be as specified in 3.3.1.5.

3.4 Performance.

3.4.1 Static overload. Both type trucks shall be capable of carrying a static load equal to 200 percent of the rated capacity of the truck with load equally distributed over the entire deck top surface without any evidence of permanent deformation, weld failure, rivet failure or failure to any part of the truck. The truck shall be tested as specified in 4.4.2 to determine conformance with the above requirement.

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3.4.2 Shock and impact load. Both type trucks shall be capable of withstanding shock loads consisting of 1–inch step drops and impact loads against a rigid hard wall when carrying an equally distributed load on the deck top surface equal to 125 percent of rated truck capacity and when traveling at a speed of 2 mph, minimum, without any evidence of permanent deformation, weld failure, rivet failure, bolt connection failure, or loosening of rubber bumper from frame. The truck shall be tested as specified in 4.4.3 and 4.4.4 to determine conformance with the above requirements.

3.4.3 Rolling resistance. Truck, with rated load distributed equally over the entire deck top surface, shall be capable of being towed with a force not exceeding 2 percent of the gross weight of the truck. Gross weight shall include the weight of the rated load and truck proper. The truck shall be tested as specified in 4.4.5 to determine conformance with the above requirement.

3.4.4 Push bar and sockets strength. The push bar, including sockets, shall be capable of withstanding without failure and permanent deformation, a minimum horizontal force of 150 lbs., applied at midpoint of the handle end of the push bar. This requirement shall be determined by the test specified in 4.4.6.

3.5 Finish . Unless otherwise specified (see 6.2), magnesium or aluminum alloy surfaces of the truck shall have a clean, natural finish.

3.6 Marking for identification. Marking for identification shall be in accordance with MIL–STD-130.

3.7 Workmanship. The truck shall be clean and properly assembled. Burrs and sharp edges shall be removed. No parts shall be damaged or impaired.

3.7.1 Welding. Surfaces to be welded shall be free of all surface contamination such as dirt, paint, oily matter and heavy deposits of oxide. Welded joints shall be sound, and shall not have burn holes, cracks, fractures or pits, fissures, flux deposited and shall be thoroughly fused to the base metal.

3.7.2 Riveting. Rivet holes shall be accurately made and shall have all burrs and fins from adjacent edges removed. Rivet heads shall be full, neatly made, concentric with rivet holes and in full contact with the surface of the members. Excessive upsetting of rivets to fill holes shall not be acceptable.

3.7.3 Bolting. Bolt holes shall be accurately made and adjacent edges shall have no burrs or fins to interfere with proper seating of bolt heads or nuts. Bolts and nuts shall be drawn tightly against lock washers.

3.7.4 Castings. Castings shall be free from blowholes, porosity, hard spots, shrinkage or cracks. The finish of the casting shall permit marking.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.3.1 and 4.3.2 and tested as specified in 4.4.1 thru 4.4.6. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performance in accordance with ANSI/ASQ Z1.4.

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4.3.1 Component and material inspection~~,~~ Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.2 Rubber bumper examination. The rubber bumpers shall be examined for nonmarking by drawing the bumper across a sheet of white bond paper using moderate pressure and shall be tested for hardness in accordance with Method 3021 of FED-STD-601. Any failure to conform to the nonmarking or hardness requirements in 3.2.1.1 shall be recorded as a defect. The lot size shall be expressed in units of bumpers. The sample unit shall be one bumper. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 0.

4.3.3 In-process inspection. Examination shall be made of the following operations to establish conformance with specified requirements. Whenever a nonconformance is noted, correction shall be made to the applicable process and all items processed.

a. Welding (surface preparation - see 3.7.1).
b. Riveting (see 3.7.2).
c. Bolting (see 3.7.3).

4.3.4 End item visual examination. The end item shall be examined for the defects listed in table II. The lot size shall be expressed in units of trucks. The sample unit shall be one completely assembled truck. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 0 for major defects and 0 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Aluminum or magnesium alloy surfaces not clean natural finish		X
	Fastening devices not plated		X
Design and construction and workmanship even (general).	Frame and deck not as specified	X	
	Sheet deck top surface not smooth, even surface	X	
	Sockets not located at the deck corners	X	
	Deck corners not round	X	
	Socket top surface not flushed with deck top surface	X	

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TABLE II. End item visual defects (continued)

Examine	Defect	Classification Major	Minor
Design and construction and workmanship even (general).	Socket does not act as stop or rigid support for push bar or superstructure supports	X	
	Push bar and superstructure not constructed or joined as specified	X	
	Bumper assembly not constructed and joined to truck as specified	X	
	Running gear not assembled to truck as specified	X	
	Caster mounting plates not constructed or joined as specified	X	
	Running gear not as specified	X	
	Part missing	X	
	Sharp burrs or edges	X	
	Part broken or damaged	X	
Meld or braze (when applicable)	Burn holes, cracks, fractures, or incomplete fusion	X	
	Pits, fissures, or flux deposits		X
Rivets	Rivet head does not seat properly		X
	Rivet loose		X
	Rivet head not full		X
	Rivet cracked, broken or missing	X	
Fasteners	Bolts, nuts, or washers not seated properly		X
	Bolt or nut not securely tightened		X
	Bolt or nut missing	X	
Marking Identification	Not legible, incomplete, missing, or incorrect	X	

4.3.5 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in this document. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of trucks. The sample unit shall be one completely assembled truck. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 0.

4.3.6 End item testing. The end item shall be tested for rack rigidity as specified in 4.4.1 and for resistance to impact load as specified in 4.4.4, The lot size shall be expressed in units of trucks. The sample unit shall be one completely assembled truck. The inspection level shall be S–2 and the AQL expressed in terms of" defects per hundred units, shall be 0.

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4.3.7 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the Section 5 requirements. Defects shall be scored in accordance with table III. The sample unit for type I and II trucks shall be one fully packed. The lot shall be the number of packs offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 0.

TABLE III. Packaging defects

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing, incorrect size, not of specified type, damaged, or otherwise defective
Workmanship (as applicable)	Not disassembled as specified; component not secured properly; loose strapping or wiring; arrangement of components incorrect; truck not anchored in crate; crate not closed and strapped as specified; or blocking inadequate
Contents	Number of trucks strapped together more or less than specified

4.4 Methods of inspection.

4.4.1 Rack rigidity test. The rack shall be mounted in the sockets. The truck shall be stayed from movement and a 5 pound force applied at the midpoint on the top of the rack. Any sway in excess of .750 (3/4) inch from a plumb line shall constitute a defect.

4.4.2 Static load. Truck shall be placed on a hard, level, concrete surface. A load equal to twice the rated capacity of the truck shall be distributed equally on the deck top surface. The load shall remain on the truck for a period of 1 hour, after which, the load shall be removed and truck shall then be thoroughly inspected. Any evidence of permanent deformation, weld or rivet failure or failure to any part of the truck shall constitute a defect.

4.4.3 Shock load. The truck with an equally distributed load on the platform top surface equal to 125 percent of rated truck capacity, shall be placed on a raised platform step 1.00 inch high, minimum. The truck shall then be rolled off the raised platform at a minimum speed of 2 mph onto a hard level surface. This test shall be repeated twenty times with the direction of truck travel perpendicular to the step, and twenty times with the direction of truck travel at a 45 degree angle to the step (10 each from the left and then from the right). After completion of the above test, the load shall be removed and the truck shall be inspected. Any evidence of failure of welds, rivets or bolt connections, permanent deformation or loosening of rubber bumper from the frame shall constitute a defect.

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4.4.4 Impact load. Type I truck shall have a test load equal to 125 percent of rated truck distributed equally over the deck top surface. For type II truck, the test load shall be the same, but the load shall consist of bags (cloth and canvas) filled with nonrigid material such as sand, sawdust or similar material. The bags shall be arranged in rows and tiers to completely fill the inside volume formed by the superstructure. The loaded truck shall then be propelled to a speed of 2 mph, minimum, and made to collide against a hard rigid wall. The collision shall involve the wall and each corner. The impact force shall act along the diagonal direction (corner to corner). The above test shall be repeated for each corner after which the load shall be removed and the truck shall be inspected. Any evidence of failure to welds, rivets or bolt connection, permanent deformation or loosening of rubber bumper from the frame shall constitute a defect.

4.4.5 Rolling resistance. The truck, with an equally distributed load on the deck top surface equal to rated truck capacity, shall be placed on a dry, smooth, level concrete surface. The loaded truck shall then be accelerated by means of a suitable towing device employing a properly positioned (tension) dynamometer. After the truck has attained a minimum speed of 2 mph, the maximum gage reading of the dynamometer shall be determined in interval of 25 feet **of** truck travel to determine conformance with the requirement of 3.4.3. Any reading on the dynamometer in excess of 2 percent of the gross weight of truck shall constitute a defect.

4.4.6 Push bar (and sockets) strength (type I truck). The truck, with push bar assembled to the sockets shall be placed on a hard, level surface. The wheels shall be blocked for this test. A force parallel to the floor surface shall be applied at the mid-point of the handle end of the push bar through a (tension) dynamometer for a period of 5 minutes. The force shall be as specified in 3.4.4. After the force is removed, the push bar and sockets shall be inspected to determine conformance with 3.4.4. This test shall be performed for each set of sockets. Any evidence of permanent deformation, or failure of the push bar or sockets shall constitute a defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity

6. NOTES

6.1 Intended use. The trucks described herein are primarily intended for use in military laundries; however, they are multi-purpose items and may be used in other materials handling operations. The type I truck is used for shipping clean packaged laundry and type II truck is used for receiving soiled laundry bundles..

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Type truck required (see 1.2).)
(c) When a first article is required (see 3.1).
(d) Finish if different from that specified (see 3.5)
(e) Selection of the applicable levels of preservation and packing (see 5.1 and 5.2)

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6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one truck. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.6 Subject term (key word) listing.

1. Bumper
2. Caster
3. Push bar
4. Static overload

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL

Preparing Activity:
DLA - CQ
(Project 3510-2018-001)

Review activities:
Army – CE, MD

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